5.80 TWh during the year has had a load factor of 66% ($= 5.80 \times 10^6$ MWh per 1000 MW $\times$ 8760 h y^{-1}). When nuclear power is used for *base load* (i.e. running at full power when operating), its load factor is only limited by how well the nuclear power stations perform. When nuclear power is not used for base load, the load factor depends not so much on the power station performance but rather on the local energy demand, available alternative energy sources and their price. Thus in Sweden during 1999 the annual load factor of hydro power was ~50%, thermal power ~22%, wind power ~19%, and nuclear power ~80%. The low load factors for hydro power and thermal power are due to the use of these in "follow the load" mode, whereas nuclear power was used for base load and wind power operated at its momentary capacity. The rather low value for thermal power is caused by its cost of operation, which is higher than the other alternatives; wind power is subsidized to make it competitive. Typical world wide average yearly load factors for all power reactors weighted with production capacity are (2011 figures): PWR 81%, BWR 73%, PHWR 71%, and RBMK 79%. These figures do not include reactors of less than 150 MWe or data from utilities with less than 4 operating reactors.

A better picture of the power station performance in situations where the demand may be less than the installed capacity is given by the *forced outage rate* which is the ratio between the number of hours a power station has been shut down because of malfunction and the total number of hours during that period. A power station which has been shut off for one month in January for repair of the main feed water pump and three months (June, July, and August) because of no demand for electricity, during which time part or the fuel elements were changed, has an 8% forced outage rate (i.e. one month out of twelve). The forced outage rate is very similar for all power stations of similar size and age. Values for new 1000 MW_e BWR or PWR stations vary from 15% up; a typical average value is 25%, normally decreasing with time.

19.15. REACTOR SAFETY

Greater than 95% of all the operation shutdowns in nuclear power stations are due to failures common to conventional thermal power stations. However, in this text we consider only safety with regard to the nuclear steam supply system which account for the other 5%. We can distinguish between three safety (or risk) levels: (i) small deficiencies due to imperfect technique, which are inevitable in any human enterprise (e.g. small "chronic" releases of radioactivity to the environment), (ii) accidents, which normally should not occur with good equipment, but still are "probable" (e.g. pipe breaks, fires, etc.) and for which protective measures must be included in the reactor design (design basis accident, DBA), and (iii) maximum credible accident (MCA) which is the most serious reactor accident that can reasonably be imagined from any adverse combination of equipment malfunction, operating errors, and other foreseeable causes.

The distinction between DBA and MCA is successively vanishing, as more possible accidents are being included in the DBA. Radioactivity releases from past reactor accidents are discussed in §13.10.2 and Chapter 22.

The broad functions of the safety systems are common to most reactors. In the event of an abnormal condition they should shut down the reactor, ensure a sufficient supply of coolant for the fuel, and contain any fission products which might escape from the fuel elements. Such safety features can be *active* (requiring some action from a control system, involving mechanical devices, and relying on an external power source in order to operate) or *passive* (built-in physical fail-safe features whose operation is not dependent on any control system, mechanical device or external power source).

Active safety systems are normally used and must be designed with a high degree of *redundance* (duplication) and *diversity* (difference in principle), so that if one safety systems fails another shall function. Several redundant active safety systems are required which are logically and physically separated from one another, and from the reactor process systems, as much as possible. They are also often based on different principles in order to reduce the probability of common-mode failures. This confers considerable immunity against events such as external explosions or internal fires. Each safety system alone should be capable of protecting the reactor core and building from further damage.

Passive safety systems are less common than active ones. However, introduction of passive (inherent) safety functions into new reactor designs is much discussed within the nuclear industry. Several ideas have been developed into new design proposals. As an example, the SECURE district heating reactor, developed by ABB-Atom[6], uses a hydro-statically metastable operating condition which causes shut-down of the reactor as soon as the normal coolant flow becomes too low for safe operation or if the core generates too much heat. Figure 19.14 illustrates the operating principle. During normal operation (a in Fig. 19.14) temperature and pressure differences keeps the system in a hydrostatically metastable state with the hot primary coolant separated from the borated shut-down solution. Power is regulated by changing the concentration of boron in the primary coolant. If either temperature in the core increases or flow through the core decreases, the gas bubble above the core will escape and the reactor shuts itself down as cold borated water enters the primary system (b in Fig. 19.14). Cooling is then secured by natural circulation inside the concrete vessel. The same principle is also proposed to be used in an inherently safe pressurized water reactor (PIUS, ABB-Atom), but many other proposed designs exist. In general, reactors with passive safety have to be designed for operation at a lower power level than reactors with active safety features.

The main threat to a reactor would involve the cooling capacity becoming insufficient because of a sudden power excursion of the reactor, a blockage of the cooling

[6] ABB-Atom AB is now owned by Westinghouse and renamed Westinghouse-Atom AB.

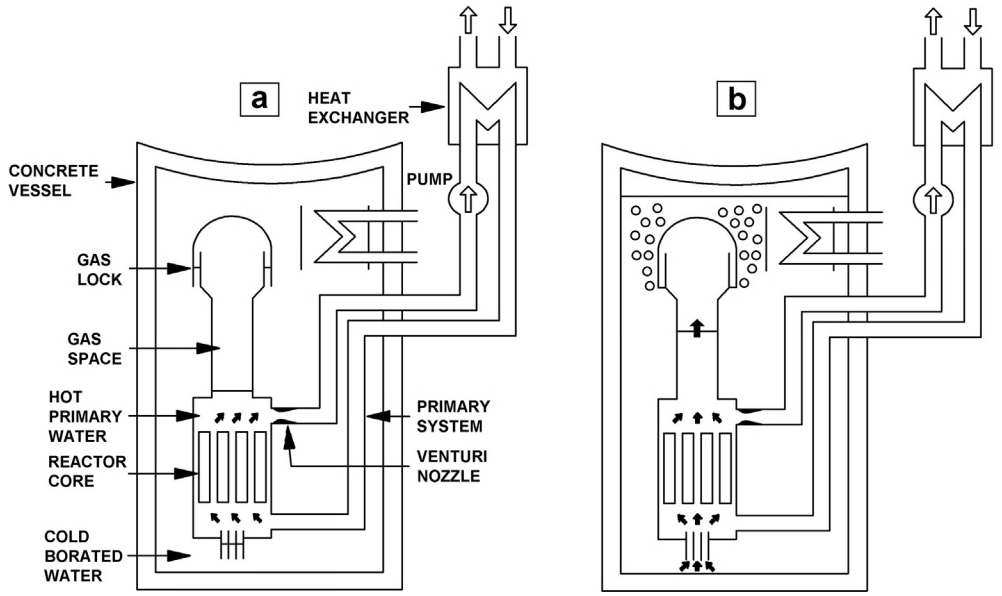


Figure 19.14 Working principle of ABB-Atom SECURE inherently safe district heating reactor.

circuit, or a loss of coolant. In either case the core could become overheated and a core meltdown might begin.

The guarantee against a power excursion is a negative temperature coefficient. For example, if the design parameters of a reactor give a maximum available $k_{\text{eff}} = 1.041$, and the temperature coefficient is $-4.2 \times 10^{-3}\,\%\Delta k/k\,^{\circ}\text{C}$, a temperature increase of 1000°C would be required to reduce k_{eff} to 0.999, i.e. to automatically stop the fission chain reaction without any of the safety systems operating. Such a reactor would also be prompt critical, as pointed out in §19.8. This is not acceptable, so at starting conditions the maximum available k_{eff} is made <1.0065 by introducing burnable poisons and control rods. After some time fission products with high neutron capture cross sections are formed (Xe, Sm, etc.), reducing k_{eff}. With Δk 0.0065, a temperature increase of 150°C would reduce k_{eff} to <1 for the example given. This can be (and is) used for reactor self-regulation. The main contribution to the negative temperature coefficient in thermal reactors comes from the decreased moderator density (and also from increased void in a BWR) with increasing temperature.

Three types of control rod are used: (*control*) *rods* for regulating small power fluctuations, *shim rods* for coarse adjustments of k_{eff}, and *scram rods* for suddenly shutting down the reactor. The scram rods can move very fast and are actuated by gravity, spring release, compressed air, etc.; a typical value may be 3 s for complete shutdown. Since the reactor time constant is much larger, a neutron excursion is effectively stopped. The number of control rods is large (≥ 100) in order to facilitate an even power density in the

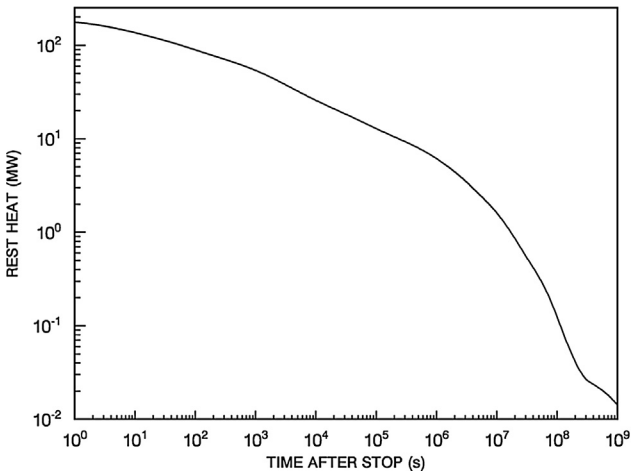


Figure 19.15 Decay heat from a 3000 MW_{th} LWR core, which has been operated continually for 300 days, as function of time after shutdown.

reactor, but the number of scram rods may be <10. Boric acid dissolved in the primary coolant is normally used in PWRs for coarse reactivity adjustment (chemical shim control). In light water reactors (BWRs as well as PWRs) injection of boric acid into the core acts as an additional emergency shutdown feature ("chemical scram").

The power of a reactor does not go to zero when it is shut off. Figure 19.15 shows the heat power due to fission product decay (logarithmic time scale) after a scram. For a 1000 MW_e (3000 MW_{th}) power reactor the rest (or after) power (or heat) is 2% of the full power effect (i.e. ~60 MW_{th}) after 1000 s. If this heat is not removed, the fuel and cladding would reach high temperatures. If nothing else would happen, the rather rapidly declining heat generation (Fig. 19.15), and the considerable heat capacity of the core, would probably limit the fuel temperature to a value below the melting point where the heat lost by thermal radiation and by convection in the remaining steam balances the decay heat. Unfortunately chemical reactions interfere. At normal cladding operation temperatures oxidation of zircaloy is very slow because of a protective oxide layer. However, at temperatures above ~900°C the protective oxide layer fails permitting the rapid exothermic reaction between zirconium and steam according to

$$\begin{gathered} \mathrm{Zr(s)} + 2\,\mathrm{H_2O(g)} \;\rightarrow\; \mathrm{ZrO_2(s)} + 2\,\mathrm{H_2(g)} \\ \Delta H = -6.67 + 2.57 \times 10^{-4}\,\mathrm{T\,(MJ/kg\,Zr)} \end{gathered} \tag{19.44}$$

where T is the cladding temperature (°C). If supplied with sufficient steam, this run-away chemical reaction can rapidly heat up the fuel to its melting point resulting in a partial core meltdown. Hot molten fuel would then accumulate on the bottom of the reactor vessel. After some time the melt could penetrate the vessel bottom at the weakest spots; pouring molten core material into the containment.

The *design basis accident* is usually considered to start with a *loss of coolant accident* (LOCA), e.g. as a guillotine break in a main recirculation line. In LWRs this immediately leads to $k_{\text{eff}} \ll 1$ because of reduced neutron moderation, independent of the position of the scram rods. However, the rest power would continue to heat the core, so core cooling is required in order to prevent a meltdown by keeping the cladding temperature below the critical range. Initiated by a lack of cooling, the heat from zircaloy oxidation caused a partial core melt-down in one of the Three Mile Island PWR reactors, Harrisburg, in 1979, but all molten material remained inside the pressure vessel. This accident had a large political impact in many countries and led to increased requirements with regard to reactor safety, e.g. improved emergency operating instructions and filtered containment venting systems.

Figures 19.16 show the *emergency cooling systems* in a BWR. There are two such systems for the core – the *core spray* and the *head spray* cooling systems (Fig. 19.16). In a modern BWR these two systems have a fourfold redundancy as Figure 19.16 shows. The systems are located in four separate zones 90° apart around the reactor and at different levels above ground. The cooling water would be taken from the wet well below the reactor. It is assumed that this well would not break; such an accident could only be initiated by an earthquake of unanticipated magnitude. In addition to the high pressure core spray there is a *low pressure coolant injection*. Further protection, especially against fission products leaking out of a damaged reactor vessel, is provided by the *containment spray*. Steam produced in the cooling is blown from the dry well into the wet well

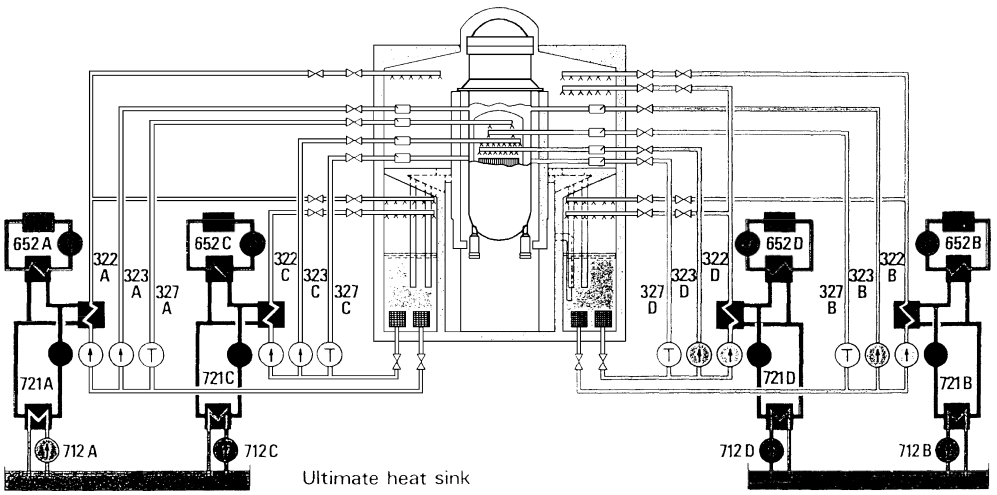


Figure 19.16 Emergency cooling and pressure relief system of ABB-Atom boiling water reactor (only one half of the cooling system is shown).

(the "blow-down" system). The water in the wet well is cooled by independent external cooling systems. The emergency cooling systems are much the same for BWRs and PWRs. In some plants mobile pumps can be connected from outside the containment building or water can be diverted from the fuel storage pool, thus providing an extra source of cooling water to the emergency systems.

A nuclear explosion in a reactor is physically impossible because of the large t_{per}. Rupture of the reactor vessel or pressure tubes due to overpressure is also highly improbable because of a number of safety valves (not shown in any of the Figures). The steam produced can be dumped directly into the condenser or into the containment in case of emergency. All piping into the core has isolation valves, which can be closed.

The reactor vessel is surrounded by several physical barriers. These barriers protect the surroundings from a core accident and protect the core from damage caused by external effects. In a PWR these barriers are the reactor tank (about 15−25 cm steel), the biological shield (1.5−3 m of concrete), and the outer containment. The whole primary circuit, including pressurizer dome, steam generator, and connecting pipes, is surrounded by concrete.

All nuclear power reactors outside the previous USSR and CMEA countries have a *reactor containment building*, the purpose of which is to contain steam and released radioactivities in case of a severe accident, and to protect the reactor from external damage. The containment is designed (and tested) to withstand the internal pressure from a release of the water in the entire primary cooling circuit (and in the case of PWRs of the additional loss of one of the steam generators), corresponding to excess pressures of 0.4 MPa. The containment is provided with a spray, which cools and condenses the steam released and washes out radioactive contaminations. The steam produced in a BWR can also be dumped directly into the condenser in case of emergency. All piping into the core has insulation valves, which can be closed if desired. Modern containments are multi-layered pre-stressed concrete and steel and can withstand the impact of a jumbo jet aircraft.

A special hazard, which is taken into account in the emergency systems, is that the cooling would be actuated too late, i.e. after a core meltdown has begun. In practice the cooling system must start $\leq$1 min after a LOCA has occurred. If the core is very hot, a violent steam and hydrogen formation could occur when the cooling water contacts the core possibly causing further heat generation by zircaloy oxidation. In this case the blow-down system, the containment, and the spray systems are still expected to contain the early energy excursion.

Under accident conditions, hydrogen might be produced from steam (or water) and zircaloy; the risk for a fire or an explosion then depends on available oxygen and mixing.

As PWR containments usually contain air, a violent hydrogen explosion could occur which would represent some hazard to a PWR with a large dry containment and even more so in case of smaller dry containments where ice is stored to be used for condensing

escaping steam (ice condensers). Such containments therefore contain igniters in order to cause a controlled hydrogen burning.

In case of inerted PS-type containments, a large hydrogen production could still interfere with the blow-down process causing pressures exceeding the design limit. To prevent breaking the containment in case of overpressure all types of containments in many countries now connect by rupture disks or valves to *filtered venting systems* designed to capture most of the vented radioactive material.

In case of any event that clearly deviates form normal behavior of a reactor (ranging from observations with safety implications to catastrophes), the event should be reported to IAEA and the appropriate organizations in other countries. For reports to the press and to the general public of safety related events in nuclear power plants IAEA and OECD/NEA recommend the INES scale, see Table 19.6.

19.16. RADIOACTIVE REACTOR WASTE

In the "ideal" nuclear reactor all fission products and actinides produced are contained in the fuel elements. In all practical reactors there are four processes through which radioactivity leaves the reactor vessel; in all cases the carrier of activity is the coolant:

(i) induced radioactivity in the cooling medium (and moderator if separate);
(ii) corrosion products containing induced activities from construction materials;
(iii) leaked fission products and actinides from faulty fuel elements;
(iv) fission products of actinides deposited on surfaces in the core ("tramp uranium").

In all these respects pressurized and boiling water reactors present some problems of concern.

Although stringent efforts are made to contain all radioactivity formed by the nuclear reactions, it is unavoidable that small releases occur to the environment. Such releases, gaseous as well as liquid, are carefully monitored by measuring stations at the plant and at some locations around the plant. The authorities (nuclear regulatory commission, nuclear power inspectorate, radiation health board, etc.) set limits for such releases. Table 19.7 gives release values for a number of different reactor types and times, as reported by the United Nations. It should be noted that these are actual releases at that time. Usually the measured releases are a small fraction of the allowed limits and tend to decrease with time as more stringent limits are imposed by the regulatory bodies, improved equipment installed or better procedures adopted.

It is important to realize that the legally *permitted release* in fact sets the normal level of releases. Very much lower releases than permitted usually means increased operating costs. Hence, economic and safety considerations sets the practical release level to some chosen fraction of the legal limit, e.g. 1% or 0.1%. A 1% practical limit would give an ample margin to the legal limit, thereby reducing the risk that accidentally increased releases exceed the legal limit (which might have dire economic consequences for the plant operator).